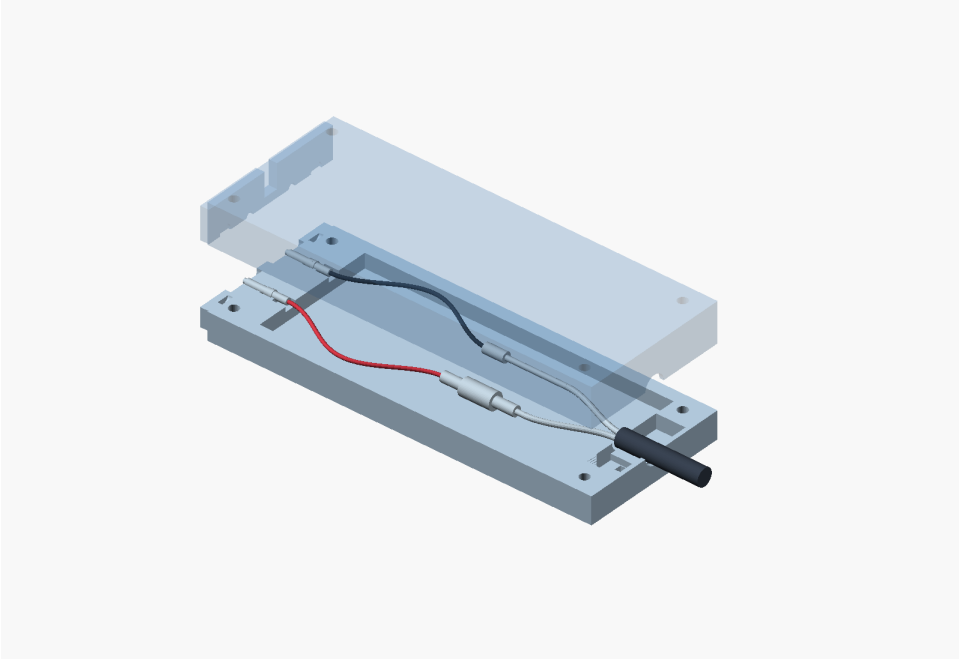
CT-DOCK-001-B | Charging-wire movement / r14

GEOMETRY STUDY — fixed-end restraint, contact holding and physical qualification remain open

External dock: 94 x 36 x 16mm

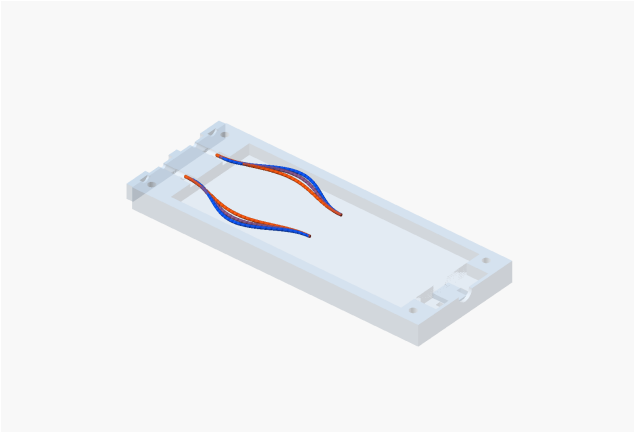


Wearable, PCB, seals, key and contact seats are unchanged.

USB red → fuse → Y=-6 → J3.1 /VIN_5V

USB black → Y=+6 → J3.2 /GND

Constant-length wire family



Free span at 23C	40.95 +/- 0.10mm per wire
In-service span bound	40.775 to 41.125mm
Conservative contact travel	2.1421mm
Minimum remaining slack	0.3858mm
Continuous radius bound	8.6961mm (required 8.382mm)
Full sweep / housing	Empty CAD intersection
Old housing control	Interference detected

The complete calculated family fits. Three colored poses illustrate it; interval bounds establish the continuous envelope.

This does not constrain a free wire to that shape. A fixed-end clamp, forming process and cyclic-flex tests are still required.

40.95mm excludes stripped conductor, crimp insertion and splice overlap. Do not use it as a production cut length.

The r05I PCB retains both recorded 0 Danger / 0 Warning / 455 Good JLC runs. No unchanged-byte DFM was repeated.

Dock retention, internal J3 wiring, pack approval, firmware and physical powered/RF/ingress tests remain open.

This dry indoor charging prototype is not an order release. IP68 is a target, not a measured or certified result.